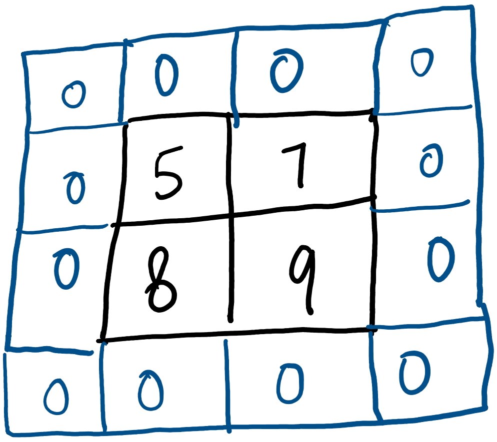
WHAT YOU WILL STUDY IN TODAY VIDEO ?

- ► What is Padding?
- ► How Padding Works and Its Use
- ► Types of Padding

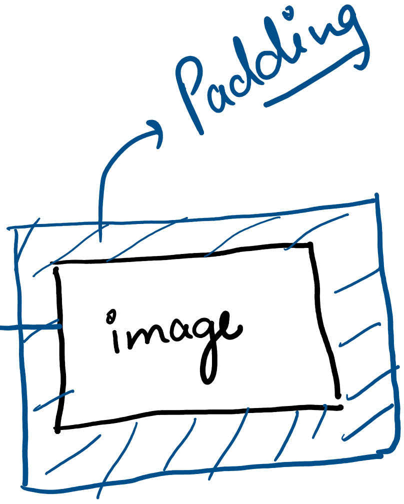
Padding

Padding=1



4x4

empty/zero pixel layer



2x2 $\xrightarrow{\text{Padding}}$ 4x4

Main Use

1. Preserve spatial information
2. More accurate model but computational power ↑

Example Assume

Input image (5×5)
Kernel (3×3)
Stride = 1

Without Padding

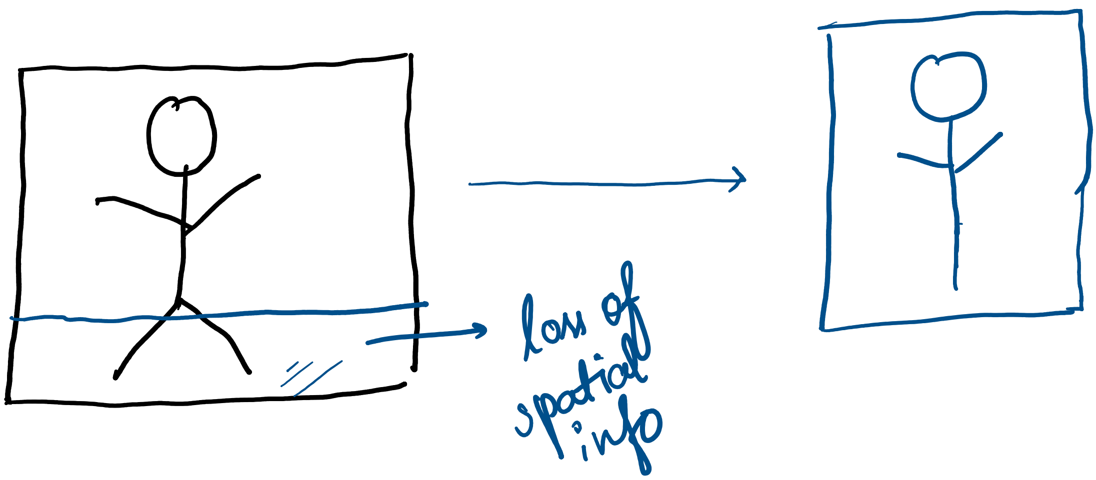
$$\text{O/P image size after convolution} = \frac{\text{Inp size} - \text{Kernel size}}{\text{Stride}} + 1$$

$$= \frac{5-3}{1} + 1$$

$$= 3 \rightarrow 3 \times 3$$

Inp image (5×5) ⟶ O/P image (3×3)

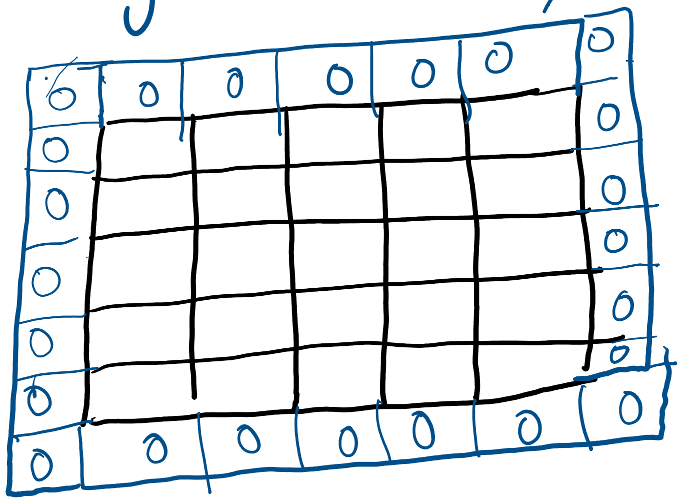
(spatial info loss)



With padding Padding = 1 ⟶ 7X7

With Padding

Padding = 1



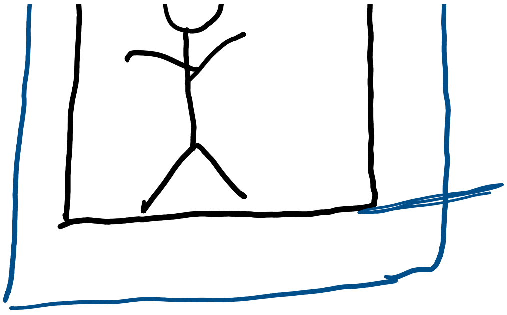
7X7 input image

$$\text{O/P size after convolution} = \frac{7-3}{1} + 1$$

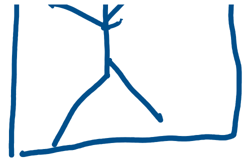
$$= \underline{\underline{5}}. \longrightarrow \underline{\underline{5 \times 5}}.$$

$$\text{Inp image } (5\times5) \xrightarrow{\text{Padding}} 7\times7 \xrightarrow{\text{Conv}} \underline{\underline{5\times5}}.$$

no spatial data loss



no loss of information



Types of Padding

Same / Zero Padding

Don't lose any spatial info

$5 \times 5 \longrightarrow 7 \times 7 \longrightarrow 5 \times 5$

When your main focus is accuracy

ie ResNet, MobileNet

Valid / No Padding

Lose of spatial info

$5 \times 5 \longrightarrow 3 \times 3$

When you main focus is speed & resource conservation